Ques	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans	B	C	A	A	B	D	D	C	C	C	B	D	D	A	B	C	C	B	B	A

Q2 : (10 marks)

A drive system has the following parameters:
$J=20kg.m^2$, $T=15+0.05N$, $T_L=5+0.06N$, where N is speed in rpm.
Initially the motor is working in steady-state, now the drive is braked by electrical braking, motor torque (T) in braking is given by $T=-10-0.04N$. Calculate time taken by the drive to stop.

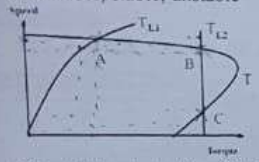
Electric Drive **Midterm Exam-** 30 Point **Instructor:** Prof. Mahdi Alshamasin **date:**24-11-2024
Time of Exam: one class hour **Student Nam**

Q1: (20 marks)

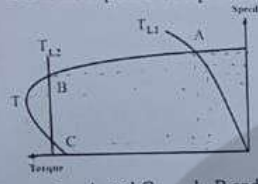
Answer the following questions by filling in the correct answer in the tabl[illegible]ed at the end of these questions

1.Stabilities for points A,B,C respectively in the figure below are

a. stable, unstable, unstable
b. stable, stable, unstable
c. unstable, stable, unstable
d. unstable, stable, stable



2.Unstable points of equilibrium in the following figure are



a. A and C b. B and C c. A and B d. A only

3.If the torque developed by the motor is $T = T_L \pm J\left(\frac{d\omega}{dt}\right) + B\omega$, the coefficient B denotes to

a. viscous friction b. Coulomb friction c. static friction d. updated viscous friction

4. In the electric drive system, the main function of the converter is

a. providing the motor with the adjustable voltage, current and/or frequency
b. matching the motor and controller to meet the load requirements
c. adapting to the changes of the environment d. increasing the torque and reducing the speed

5. If load is driven by two motors, the drive classification is

a. main and auxiliary b. interconnected c. group d. regulated

6. The electric drive system with following motor torque and load torque expressions is stable

a. $T_m=10$, $T_L=-2N$
b. $T_m=2+2\omega^2$, $T_L=25+5\omega$
c. $Tm=10$, $T_L=5\omega+1$
d. $Tm=10$, $T_L=10-2\omega$

7. The reactive (passive) load torque industrial machinery is

a. crane b. winch c. hoisting mechanism d. turning machine

8. When the transient time of transient operation of EDS is small, the right statement is

a. static performance is good b. static performance is poor
c. dynamic performance is good d. dynamic performance is poor

9. Motion equation for steady-state operation of linear electric motor is

a. $F_m - F_l = v\frac{dm}{dt}$
b. $F_m - F_l = m\frac{dv}{dt}$
c. $T - T_L = J\frac{d\theta}{dt}$
d. $T - T_L = J\frac{d^2\theta}{dt^2}$

10. Hydraulic pump torque-speed characteristic is

a. linear rising characteristic b. non-linear rising characteristic
c. independent of speed d. linear decreasing characteristic

11. Positive -displacement compressor torque-speed characteristic is

a. non-linear decreasing characteristic b. independent of speed
c. quadratic characteristic d. linear decreasing characteristic

12. The crane motor lifts a load of 20 N at a speed of 1 m/s using a drum with a radius of 0.5 m. If the system efficiency equals 0.8, the referred load torque to motor shaft is:

a. 25 N.m b. 16 N.m c. 12.5 N.m d. 2.5 N.m

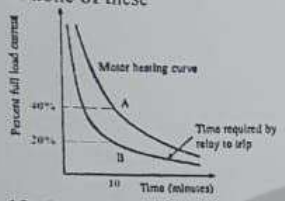
13. For load equalization, the true statements are
a. peak current drawn by the motor from the source is reduced by a large amount.
b. fluctuations in motor torque and speed are reduced c. In variable speed and reversible drive, a flywheel cannot be mounted on the motor shaft d. all of these are correct

14. Electric motor has moment of inertia of 2 kg.m^2, connected through gearbox of 10 gear ratio to drum with moment of inertia 5 kg.m^2. The drive system total (equivalent) moment of inertia equals
a. 2.05 kg.m^2 b. 2.08 kg.m^2 c. 1.05 kg.m^2 d. 0.05 kg.m^2

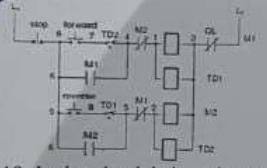
15. In four quadrant operation of a motor driving a hoist load (elevator), if the cage is unloaded and the motor speed is downward, then the motor is working in
a. the first quadrant b. the third quadrant c. the fourth quadrant d. the second quadrant

16.If the motor works in the fourth quadrant. This mode of operation is a
a. forward motoring b. reverse motoring c. reverse braking d. forward braking

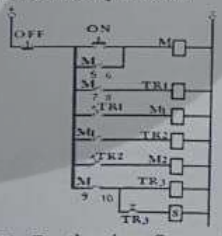
17. In the figure below, what is the correct statement?
a. through the first10 Mins of motor working, the motor can be overloaded by 20% up to 40%
b. if the load exceeds the rated value by 40%, the thermal relay will disconnect the motor within 10 Mins.
c. if the load exceeds the rated value by 20% or 40% , the thermal relay will disconnect the motor within 10 Mins.
d. none of these



18. Control circuit below was used to in 3-phase motor, the function of the NC contacts (M1 and M2) is:
a. to protect the motor from short circuits. b. to ensure the proper functioning of both timer relays together
c. to prevent short circuit in the control circuit d. to prevent short circuit in the three-phase power supply



19. In the circuit below, the contactor (s) is used for
a. overload protection b. motor plugging braking c. motor dynamic braking d. motor regenerative braking



20. For the given figures of AC motor stator connections, if we start the motor using a Y-connection, the source (line) current will decrease by a factor of
a. $\sqrt{3}$ times b. 3 times c. $3\sqrt{3}$ times d. $\frac{1}{\sqrt{3}}$ times

